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RESEARCH-ARTICLE

Optimizing Uncertainty Estimation on Scientific Visualizations Using Learning Models

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Abstract—Scientific visualizations (SciVis) translate numerical and spatial data as images, enabling scientists to better understand the phenomena described by the data and gain insights that may be overlooked by statistical methods alone. In recent years, machine learning, particularly Deep Learning (DL), has driven significant advancements in SciVis. By combining classical numerical techniques with DL methods, we can create models that potentially achieve the accuracies needed for real-world industrial applications and simulations. However, DL models often exhibit undue confidence, particularly when faced with inputs that differ significantly from the data they were trained on, known as out-of-distribution (OOD) inputs. This can lead to misclassifications where the model still produces high confidence scores, reflected in elevated softmax output values, even though its prediction is incorrect. Ideally, in such cases, the model should indicate uncertainty, effectively saying, “I do not know.” In this work, we enhance the trustworthiness of DL models by quantifying their uncertainty and enabling selective classification, where the model abstains from making predictions when uncertainty is high. Instead of providing a single prediction, our model outputs a prediction distribution, allowing users to determine whether human intervention or other actions are necessary for handling high-uncertainty outputs. This paper evaluates uncertainty quantification across three different tasks: predicting airfoil pressure and velocity using a Reynolds-Averaged Navier-Stokes (RANS) model on a custom airfoil dataset, image classification using the ImageNet1K dataset, and digit recognition using the MNIST handwritten digits dataset.

I. INTRODUCTION

Uncertainty evaluation is critical for some DL tasks. For instance, in pedestrian detection for self-driving cars [1], an AI model may not recognize a human who is partially obscured by a bicycle. Evaluating uncertainty is essential in any scenario that is novel compared to the training data. For example, in [2], researchers trained a path-planning robot that can ask for help when it does not understand a user’s command by incorporating uncertainty measurements into the learning model. Similarly, in [3], the authors bench marked one point

wise model and three approximate Bayesian DL models to predict uncertainty in cancer classification using three RNA-seq datasets. These examples illustrate the ability to output “I do not know” and seek assistance is vital for the reliability of DL models.

Numerous techniques have been developed to quantify uncertainties in DL. For example, temperature scaling [4] is a post-processing technique to calibrate neural networks. However, this method requires learning the scaling parameters during DL training, which can be limiting. Bayesian Neural Networks (BNN) [5] extend standard networks with posterior inference, allowing for learning a probability distribution over outputs, where everything – including model parameters (weights and biases) – has a probability distribution attached. BNNs mitigate over fitting, enable learning from small datasets, and provide insights into prediction uncertainty, although they are computationally intensive. In [6], researchers introduce Peer Loss Functions, an approach that enables models to learn even from noisy labels, which is particularly useful in noisy environments. Evidential Learning [7], [8] replaces the traditional DL softmax layer with an evidence vector and learns the Dirichlet [9] distribution parameters to handle uncertainty. In paper [10], a multi-input multi-output (MIMO) configuration is employed to generate multiple predictions from a single model’s forward pass, training multiple subnetworks that independently learn the task. One of the most successful methods in evaluating uncertainty is ensemble learning [11], where multiple model’s are trained independently on identical data, and their outputs are aggregated. Since ensemble learning is computationally and memory intensive, research has focused on making it more efficient. For instance, in [12], a snapshot ensemble with cyclic learning rates is proposed, where snapshots of the model’s weights are saved periodically during training and used to form an ensemble. Another approach, detailed in [13], involves distillation, where a single model is trained to match the output of all ensembles combined as

*These authors contributed equally to this work

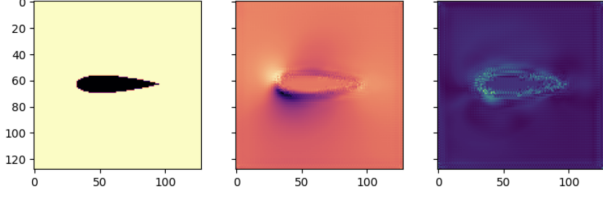


Figure 1: Simulation of the pressure on an airfoil. The first image shows the initial airfoil shape, the second image shows the predicted pressure values, and the third represents the uncertainty of the model. Input image was obtained from the dataset in article [17].

one. We will describe the ensemble, evidential, MIMO, and Bayesian techniques in more detail in section II.

There is also interesting research in DL for SciVis. For instance, in [14], scientists use oneAPI [15] to compress and then decompress extensive datasets quickly enough to enable real-time, interactive renderings of SciVis models. In [16], researchers present an excellent overview of Physics-Based DL and implement a model to find the average motion and pressure distribution of a turbulent flow around an airfoil at different Reynolds numbers and angles of attack. We will be using the dataset used in this work as the basis for our SciVis experiments. Figure 1 presents an example of SciVis using DL.

The **main contributions** of our study are the introduction of different uncertainty evaluation techniques for scientific visualizations using learning algorithms. We also provide a detailed comparison of selected uncertainty estimation techniques, discussing optimal options based on available computational resources.

The paper is organized as follows. In Section II, we describe the datasets and uncertainty evaluation methods; in Section III, we summarize our results; and in Sections IV and V, we present our conclusions and future work, respectively.

II. METHODS

This section provides a detailed overview of the uncertainty estimation methods we employed in our models.

A. Ensembles

Ensembles [11] aggregate outputs from multiple models that are trained independently on the same dataset. For a given input x and label y , a neural network (NN) predicts the distribution $p_\theta(y | x)$ where θ represents the NN’s parameters. An ensemble consists of M models and their hyperparameters, $(\theta_m)_{m=1}^M$, trained in parallel. The combined prediction is given by $p(y | x) = M^{-1} \sum_{m=1}^M p_{\theta_m}(y | x, \theta_m)$. Uncertainties are calculated from the standard deviation of predicted probabilities across the ensemble. We follow the axiomatic approach from probabilistic opinion polling [18] to aggregate the results of the different ensemble members. The key axioms are:

- 1) **Unanimity Preservation:** If all P_i (the opinions of each individual i) are identical, the $P_{(P_1, \dots, P_n)}$ (collective probability) is identical to them.
- 2) **Eventwise Independence:** The collective probability of any event is dependent solely on individual probabilities.

This approach precludes denoting the collective probability for any event A as the weighted average $D(x_1, \dots, x_n) = w_1 x_1 + \dots + w_n x_n$, where $w_1, \dots, w_n$ are the weight assigned to each ensemble member. We modify the weights w_i of the ensemble based on the accuracy of individual models. To improve efficiency, we reuse the models generated during Hyper-Parameter Optimization (HPO), selecting the N best models so that no extra computational time is required to calculate the uncertainty.

B. Evidential Learning

Evidential Learning [7], [8] replaces the classical NN’s softmax layer with an evidence vector, which predicts the Dirichlet distribution parameters [9]. For a given sample x_i in a K -class classification problem, the cross-entropy loss to be minimized for learning evidence e_i is $\mathcal{L}(e_i, \theta) = \sum_{k=1}^K y_i^k (\log S_i - \log(e_i^k + 1))$; where y_i is the one-hot k -dimensional label, S is the total strength of the Dirichlet distribution $DIR(p | \alpha)$ parameterized by $\alpha \in \mathbb{R}$ and $S = \sum_{k=1}^K \alpha_k$. The evidence e_i is calculated as $e_i = g(f(x_i; \theta))$, where f as the output of the NN parameterized by θ and g as the evidence function that ensures e_k remains non-negative. Following [7], $\alpha_k = e_k + 1$. During inference, the probabilities for each class k are then given by $p_k = \frac{\alpha_k}{S}$, and the uncertainty for each sample is calculated as $u = \frac{K}{S}$. For a detailed step-by-step explanation of evidential learning with code, we recommend [19], [20] as well as the code for this project provided in Appendix.

C. Binary Evidential Learning

A limitation of standard evidential learning is that it provides a general uncertainty value rather than per-class uncertainty values. To address this, we introduced binary evidential learning in our previous work [21]. This method trains k binary classifiers using calculations similar to standard evidential learning but adapted for two classes, as illustrated in Figure 2. For example, when classifying digit 1, one class will be “yes” if the image contains a 1, and “no” for any other digit. While this approach increases computational complexity of the model, it offers more detailed uncertainty analysis, which we explore further in the results section.

D. MIMO

We also implement the uncertainty estimator described in [10], which uses a multi-input multi-output (MIMO) configuration to concurrently train multiple independent subnetworks within a single network. The advantage of this approach is that we can evaluate all subnetworks simultaneously during testing, thereby achieving the benefits of ensemble methods in a single forward pass. At test time, the same input is repeated M times, and the subnetworks make M independent predictions on this input, forming an ensemble for a single

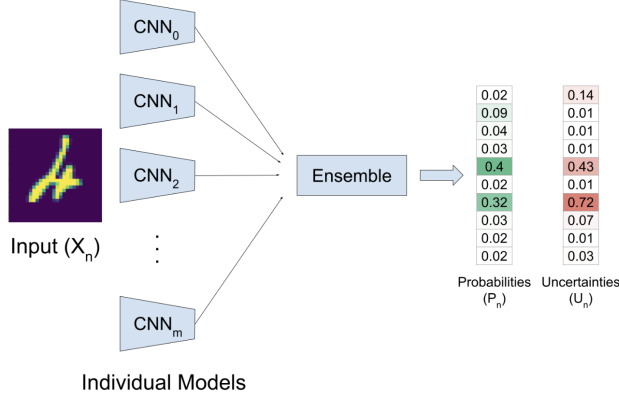


Figure 2: Binary evidential classifier diagram.

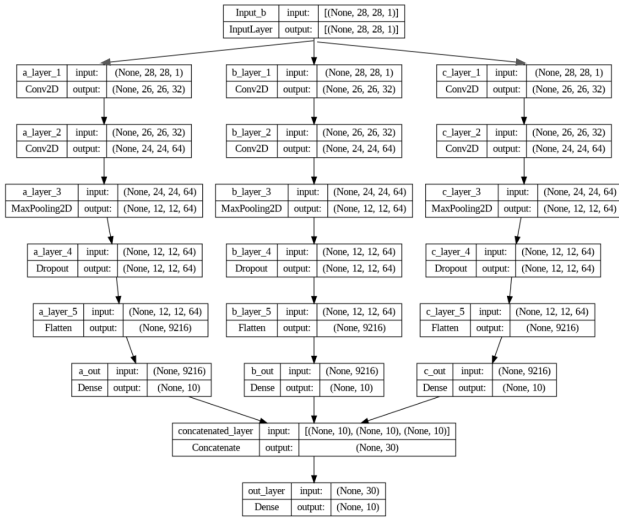


Figure 3: Multi-Input Multi-Output network for three subnetworks.

robust prediction. This approach is particularly advantageous for AI-specific architectures like [22], which are designed to train very deep models efficiently.

E. Bayesian Neural Networks

In regular supervised DL, the model makes predictions on the input x to generate an output $y \mapsto f(x, w)$ via a network f with weights w . Using a dataset D of labeled examples $D = \{x_i, y_i\}$, the model compares the generated output to the known label y_i and minimizes a loss function $L(D, w)$ with respect to the weights w . $L(D, w) = \sum_{x_i, y_i \in D} (y_i - f(x_i, w))^2 + \lambda \sum_d w_d^2$, where y_i is the target label, x_i is the input, $f(x_i, w)$ is the output, and λ is a regularization factor for large w values. The model iteratively refines its predictions, gradually improving accuracy.

In a Bayesian Neural Network (BNN) [5], each weight is represented as a distribution rather than a single value. This

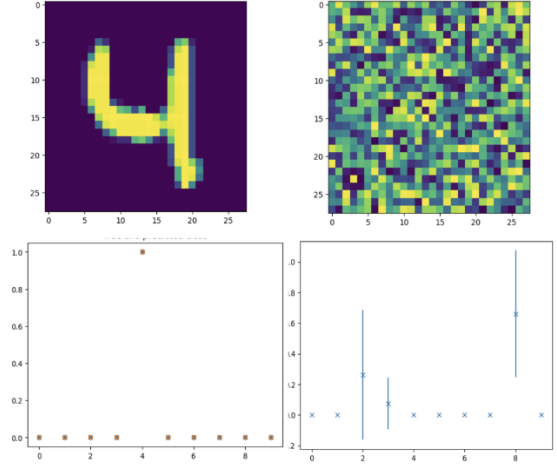


Figure 4: MNIST Handwritten Dataset. The first row presents two images: the first is a digit '4', and the second is an OOD random image. The second row displays the output label (from 0 to 9) against the softmax output. For the first image, only label 4 has a high softmax output, with no std. dev. bars, indicating high certainty in the prediction. In contrast, for the second image, the highest softmax output is for label 9, but the large std. dev. bars indicate significant uncertainty, highlighting the model's uncertainty in this prediction.

means each neuron considers a range of weight values for each input, adding a layer of probabilistic reasoning by outputting a probability distribution that reflects uncertainty. While BNNs are very computationally intensive, researchers have shown that using dropout (during training and inference) can be interpreted as a Bayesian approximation of the Gaussian process [5]. We applied this approximation to perform a BNN uncertainty estimation in our implementation.

For more information on our uncertainty estimation for Ensemble [11], Evidential [7], MIMO [10], and Bayesian [5] models, readers are encouraged to refer to the code provided in Appendix.

F. Models and DataSets

In our experimentation, we employed the following datasets:

- **MNIST Dataset** [24]: We used Convolutional Neural Networks (CNN) for the classification of handwritten digits. An example of the input can be seen in Figure 4.
- **ImageNet 1K Dataset with Vision Transformers (ViT)** [25]: This dataset was used under a Projected Gradient Descent (PGD) adversarial attack [26]. An example of the input can be seen in Figure 5
- **Deep Flow Prediction** [17] and [27]: This framework predicts turbulent flow (Reynolds-averaged Navier Stokes) for airfoils using deep learning with a U-Net network [28]. A sample input can be seen in Figure 6.

Model specifics are elaborated on in Appendix (code files).

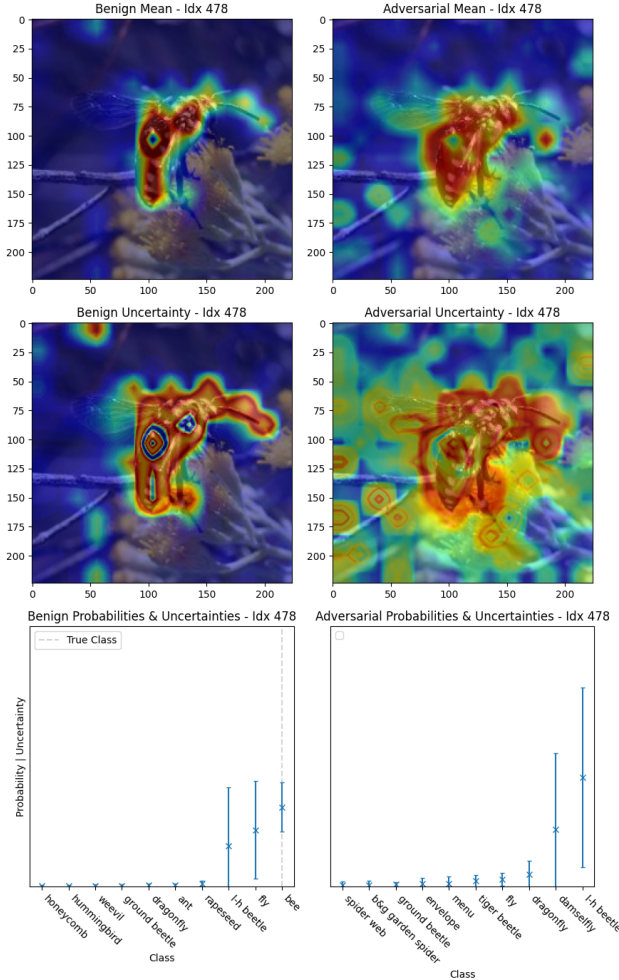


Figure 5: Grad-CAM [23], red indicates the main pixels used by the model to decide the classification. In row 1 col 1 contains the original image without manipulation, row 2 col 1 presents the uncertainty of the ensemble of the same regular image. In row 2, col 1 contains the the regular image under adversarial attack, row 2 col 2 presents the uncertainty of the image under adversarial attack. The two images on row 3 represent the error bar diagrams for the same image without (left) and with (right) adversarial attack. It can be seen that images with adversarial modification also have high uncertainty values.

G. Using Uncertainty with Selective Classification

Once the uncertainty is calculated, it can be used to withhold predictions or output an “I do not know” response if the input contains too much uncertainty. We employ a selective classification algorithm [21] that delegates predictions to human experts when uncertainty exceeds a predetermined threshold. In this approach, incorrect predictions that are corrected post-review are labeled as correct, while unnecessary deferrals are considered incorrect. This strategy discourages excessive

deferrals, promoting a balance between human expertise and automated efficiency.

For the ImageNet 1K classification problem using ViT under adversarial attack, we calculate uncertainty not just as the standard deviation of the output label, but as an aggregate standard deviation of all the different outputs for each label with a softmax output above 0. As shown in Figure 5, the image subjected to an adversarial attack exhibits high uncertainty values, making it a candidate for rejection in classification. This process improves the model’s overall accuracy and enhances safety.

III. RESULTS

Our research focuses on evaluating the efficiency and accuracy of various uncertainty quantification methods using two different systems:

- **Desktop Computer:** AMD Ryzen Threadripper 3960x and one Nvidia RTX 2080 Ti.
- **Polaris Supercomputer:** AMD EPYC processors and four NVIDIA A100 GPUs at Argonne National Laboratory. For specifics on Polaris [29].

A. Accuracy

Each uncertainty methodology (ensembles, evidential learning, binary evidential, MIMO, and Bayesian) offers unique advantages and limitations. For example, in the MNIST Handwritten Digits classification task, uncertainty estimation helps detect OOD images, such as the randomly generated one shown in Figure 4. The standard deviation bars indicate that the ensemble model consistently struggles with labeling this image. Similarly, standard deviation bars, combined with gradcam [23], are used in the ImageNet 1K dataset, as seen in Figure 5, to detect adversarial attacks by highlighting high-uncertainty areas in the images. In the Deep Flow Prediction task [17], where the goal is to predict pressure and velocity(x,y) on an airfoil using the Reynolds-averaged Navier-Stokes numerical method as the theoretical target, the results of uncertainty evaluations are visualized in Figure 6. The figure shows the output from different ensemble models, the aggregated results, and the corresponding uncertainty visualization. Uncertainty, in this context, represents the areas where the model has difficulty making consistent predictions. Figure 7 summarizes the overall uncertainty results for the three outputs, indicating that the x-component of velocity has more uncertainty than pressure and velocity in the y direction.*

Table I presents a summary of the accuracy’s achieved with different uncertainty estimation methods across various datasets. The table uses the following codes: NoUnc.= Model without uncertainty estimation, UncEns. = Uncertainty estimation using ensembles, UncEvi. = Uncertainty estimation using evidential learning, UncMIM = Uncertainty estimation using MIMO, UncBay. = Uncertainty estimation using BNN. All models maintained the same architecture and hyperparameters,

*The Hyperparameters for the models are all the same (except for ensembles that create their own hyperparameters).

with eight ensembles used for ensemble estimation and eight subnetworks for MIMO. For the MNIST dataset, where the baseline model accuracy is very high, adding uncertainty estimation generally reduces accuracy due to some images being flagged with high uncertainty and not evaluated. For ensemble learning, which slightly improves accuracy, results are consistent with expectations, as ensembles combine multiple high-accuracy models and are known for providing robust uncertainty estimates. The same trend is observed in the other two datasets, with ensembles consistently providing better accuracy. While evidential learning performs slightly worse in accuracy but it does much better in execution time.

Dataset	NoUnc.	UncEns.	UncEvi.	UncMIM	UncBay.
MNIST	0.988	0.991	0.964	0.981	0.969
Imagenet 1K	0.820	0.840	0.800	-	*
Airfoils	0.972	0.979	0.89	0.976	0.976

Table I: Datasets accuracy for different uncertainty evaluations. Accuracy calculated as dividing the number of correct predictions by the total number of predictions.

B. Performance

This section presents the execution times for different datasets and uncertainty estimation methods on the two hardware platforms described at start of this section. We used MPI [30] ranks to distribute the ensemble models across different CPU cores on desktop computers with a single GPU and across nodes on the Polaris supercomputer. Since the individual ensembles operate independently, they can be distributed without concerns about race conditions or data dependencies.

As shown in Table II, ensemble learning scales efficiently with the number of nodes and GPUs, making it the preferred uncertainty estimation method for users with access to a supercomputer. The models can be executed in parallel, and the execution times for ensemble learning are comparable to training a standard DL model.

N.Ensembles	N.Nodes	N.GPUs	Time(sec)
4	1	4	3min17sec
4	2	8	2min27sec
8	1	4	4min35sec
8	2	8	3min14sec
16	1	4	5min58sec
16	2	8	4min55sec
16	4	16	3min25sec

Table II: MNIST dataset timings on Polaris using ensembles.

Using the MNIST dataset, Table III shows that on a desktop computer with only one node, the most computationally intensive uncertainty evaluations are MIMO, Ensemble, and Bayesian, while evidential learning is more efficient. Although

- Since Image Net 1K has one thousand different labels, creating a subnet for each is impractical.

* For ViT bayesian uncertainty was too computationally intensive and all literature points to ensemble as the most accurate way of evaluating it.

ensembles provide more robust uncertainty values, they are computationally intensive. However, due to their scalability on HPC systems, ensembles do not add significant overhead, unlike Bayesian uncertainty evaluation, which does not scale as effectively, limiting the benefits of using HPC. From a computational perspective, MIMO is slow on both desktop and HPC platforms, but on a specialized machine like the Cerebras CS-2 [22], it could be the faster method due to its ability to handle extensive layers like those produced by a MIMO model. We do observe the same execution time patterns when using the airfoils dataset as represented in Table IV. In this airfoils dataset, there is no test for Binary evidential, or MIMO, since in this case we are doing prediction and these two methods use the number of classes to classify to create the uncertainty.

Unc. Method	Lambda Hw	Polaris
No Uncertainty	1min21sec	1min01sec
10 Ensembles	17min87sec	2min15sec
Evidential	1min22sec	1min17sec
Binary Evidential	12min24sec	10min10sec(&)
MIMO (10 subnets)	39min95sec	4min45sec
Bayesian	10min55sec	2min12sec

Table III: Execution time (in sec) of different uncertainty evaluation methods on the MNIST dataset.

Unc. Method	Lambda Hw	Polaris
No Uncertainty	11min31sec	4min12sec
10 Ensembles	78min87sec	5min27sec
Evidential	7min31sec	4min12 sec
Bayesian	35min69sec	17min10sec

Table IV: Execution time (in sec) of different uncertainty evaluation methods on the Airfoil dataset.

In Figure 8 we demonstrate the scalability of the MIMO model on one node (with 4 GPUs) of Polaris. We observe that batch size plays a crucial role in keeping all 4 GPUs fully utilized.

Overall, if no supercomputer is available, evidential learning may be the most efficient option due to its lower computational demands. However, if you have access to a system like Polaris, ensemble learning is recommended due to its scalability and robust uncertainty estimation capabilities. MIMO works well for classification problems and also for some of the new artificial intelligence hardware like cerebras.

IV. CONCLUSION

This research has made significant strides in exploring uncertainty quantification in DL, a field crucial for making DL trustworthy in applications such as medical imaging, self-driving cars, SciVis, and other fields. Our findings highlight the potential acceleration benefits of employing HPC and distributed systems for uncertainty evaluations, underscoring

& Binary Evidential Learning can be easily distributed, the time in Polaris doesn't include using MPI to distributed the different binary models; with this improvement the timings should be similar to 10 Ensemble

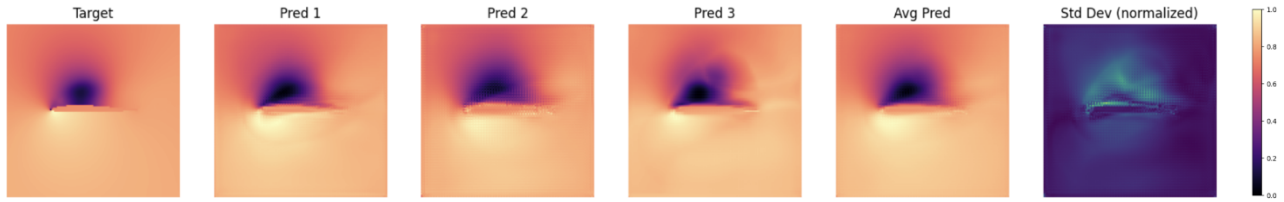


Figure 6: Target output, predictions from each ensemble Pred 0-3, aggregation of the ensembles and standard deviation/uncertainty.

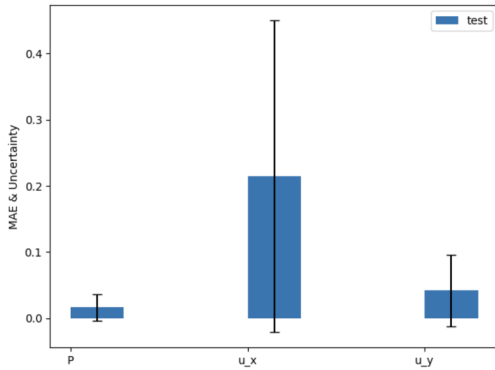


Figure 7: MAE (Mean Absolute Error) and uncertainty (as long bars) on the test set.

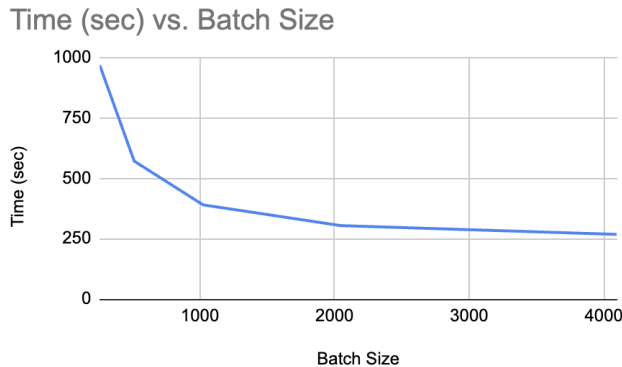


Figure 8: Uncertainty evaluation using MIMO. Scaling batch-size to the node.

the importance of these technologies in enhancing DL models. For scenarios with limited HPC resources, evidential learning emerges as a viable alternative, offering valuable uncertainty insights with minimal overhead.

We demonstrated the practical applications of uncertainty by showing how it can be used to “reject” OOD inputs, as illustrated in Figure 4. Additionally, we highlighted the value of uncertainty evaluations in cases where a ViT-based classifier misclassifies under adversarial conditions (Figure 5). Finally, we showcased the use of uncertainty evaluation in SciVis to identify areas within an image where the model encounters difficulties (Figure 6).

In evaluating the computational intensity of different methods, we found that while evidential learning is more efficient for general use, MIMO could be more suitable for specialized hardware like the Cerebras CS-2 due to its architecture. Although computational considerations are essential, the primary focus remains on the specific use case and the critical role of uncertainty in ensuring reliable model performance.

V. FUTURE WORK

For future work, we aim to enhance fluid dynamic visualization by employing a more realistic dataset (3D instead of 2D) and using a more accurate target model, such as the one obtained by using Large Eddy Simulation ([31], [32]), instead of the Reynolds-Averaged Navier-Stokes approach. Additionally, we plan to extend our uncertainty study to the acceleration of partial differential equation (PDE) solutions using DL, a method widely applied in SciVis. This will allow us to develop a more general uncertainty framework, improving the accuracy of DL models used in SciVis.

In recent years, there has been considerable research on Active Learning (AL) [33], a human-in-the-loop model that enhances task performance by selecting examples for human annotation, with uncertainty sampling as a key component. In the context of DL, AL aims to maximize model performance gain while minimizing the number of samples that require annotation. We plan to incorporate AL into our SciVis models, enabling the use of larger, unlabeled datasets and further improving model accuracy and reliability.

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APPENDIX

Access our experimental code at :
<https://github.com/epautsch/UncertaintyANL>
<https://github.com/mpantoja314/Uncertainty>

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